

TASK 02 – Prompting for Creativity

Objective

The objective of this task is to explore how prompt structure, role assignment, context, constraints, and examples influence the creativity and quality of AI-generated outputs. The task demonstrates how progressively refining prompts results in more innovative, detailed, and practical responses.

Introduction

Prompt engineering plays a significant role in creative content generation. By providing clear instructions, assigning expert roles, specifying the desired output format, and including constraints or examples, Large Language Models (LLMs) can generate more relevant, structured, and imaginative responses. This task investigates how different prompting techniques affect the generation of hackathon themes.

Prompt 1 – Basic Prompt

Prompt

Suggest a hackathon theme.

Output Summary

The AI generated a list of general hackathon themes such as:

- AI for Social Good
- Smart Cities
- Healthcare & Wellness
- ClimateTech & Sustainability
- Data Analytics & Business Intelligence

Each theme included only a brief description without objectives, problem statements, or implementation details.

Observation

The response was useful but generic. Since no audience, purpose, or output structure was specified, the AI produced broad ideas with limited detail.

Prompt 2 – Improved Prompt

Prompt

Suggest five innovative hackathon themes for college students focused on AI and sustainability.

Include:

- Objective
- Sample Problem Statements

Output Summary

The AI generated five detailed hackathon themes including:

- AI for Smart & Sustainable Campuses
- Climate Action with AI
- AI for Sustainable Agriculture
- Circular Economy & Smart Waste Management
- Green Mobility & Smart Transportation

Each theme contained:

- Clear objective
- Multiple real-world problem statements
- Bonus challenge ideas
- Judging criteria

The output became significantly more practical and suitable for organizing an actual college hackathon.

Observation

Adding the target audience, domain, and expected sections improved creativity and ensured that the generated ideas addressed real-world sustainability challenges.

Prompt 3 – Advanced Prompt (Role + Structured Prompting)

Prompt

Act as the Head of Innovation at a global technology company.

Generate five unique hackathon themes for engineering students.

For each theme include:

- Theme Name
- Vision
- Target Participants
- Problem Statements
- Judging Criteria
- Expected Impact
- Prize Categories

The ideas should be original, practical, and suitable for a 24-hour hackathon.

Output Summary

The AI produced a professional hackathon blueprint featuring themes such as:

- AI for a Sustainable Tomorrow
- Human-Centered AI
- Smart Industry 5.0
- AI Agents for Everyday Life
- Resilient Communities Through Technology

For every theme, the response included:

- Vision statement
- Target participants
- Real-world problem statements
- Judging criteria with weightage
- Expected impact
- Prize categories
- Recommended 24-hour hackathon format

The output resembled documentation prepared by a professional event organizer or innovation team.

Observation

Assigning an expert role and defining a comprehensive output structure resulted in highly creative, detailed, and actionable content suitable for real-world implementation.

Comparison Table

Feature	Prompt 1	Prompt 2	Prompt 3
Creativity	Moderate	High	Very High
Level of Detail	Low	Medium	High
Context Provided	No	Yes	Yes
Role Assignment	No	No	Yes
Output Structure	Basic	Semi-Structured	Fully Structured
Practical Application	Limited	Good	Excellent
Real-World Usability	Low	High	Very High

Prompt Engineering Techniques Used

Technique	Purpose	Impact
Context Prompting	Defined AI and sustainability domain	Generated more relevant themes
Audience Specification	Focused on engineering students	Tailored ideas to the target users
Structured Prompting	Requested specific sections	Produced organized and comprehensive outputs
Role Prompting	Assigned "Head of Innovation" persona	Increased professionalism and creativity
Constraints	Specified 24-hour hackathon	Made ideas realistic and implementable

Key Learnings

- Clear prompts significantly improve the relevance of AI-generated content.
- Providing context helps the AI understand the intended purpose.
- Role prompting influences the expertise and tone of the response.
- Structured prompts lead to organized and complete outputs.
- Constraints help generate realistic and actionable ideas.
- Prompt refinement enhances creativity while maintaining consistency.

Conclusion

This task demonstrated that creative outputs generated by Large Language Models improve substantially when prompts are refined with context, role assignment, structure, and constraints. The progression from a simple request to a professionally structured prompt resulted in responses that were more innovative, detailed, and suitable for real-world applications. It highlights the importance of prompt engineering in maximizing the effectiveness of AI for creative problem-solving and content generation.

Tanushri Mahesh Palleda,
Prompt Engineering Intern,
SkillCraft Technology.